Financial Document QA on FinanceBench

Live demo: finqa-demo.vercel.app (step-by-step replay on sample questions, and every answer with its cited pages). Code and full results: github.com/moriowen/finqa. The system diagram is on page 3.

This is a comparative experiment, not a product. It compares Qwen3.5 at 2B, 9B and 27B, each with and without a checking pipeline, to decide which model and setup to use later. One rule shaped the design: a wrong answer is worse than no answer. If a configuration cannot point to evidence that code has checked, it refuses and gives the reason. Every configuration is reported as three counts: correct, wrong and refused.

Terms used

Term	Meaning
Bare	The model gets the question and the retrieved pages in one prompt and writes an answer. Nothing checks it.
Full pipeline	The same model and pages inside the checking pipeline of section 1. Any failed check ends in a refusal.
Refused	No answer is given (NOT_FOUND or AMBIGUOUS), with the reason. Counted apart from wrong answers.
Provenance	The filing, PDF page, exact quote and surrounding passage behind each value in an answer.
Statement pack	The pages of a filing that are its primary statements: income statement, balance sheet, cash flow statement.
BM25	Keyword ranking: orders pages by the question's words they contain, rare words counting more.
Embedding search	Ranks pages by similarity of meaning, using one vector per page and one per question.
Gold answer, gold page	The answer and evidence page written by FinanceBench's analysts. Used only for grading, never as input.
FP8	The 27B model's weights stored in 8 bits instead of 16, so that it fits on one GPU.

1. System design

Parsing. PyMuPDF reads each of the 363 filings and stores every page twice: plain text for search, and layout text that keeps table rows and columns (joined with "|") for the model to read. One chunk is one PDF page, so tables stay whole and the page number is the provenance. At index time, code marks the pages whose title is a primary statement (income statement, balance sheet, cash flow statement). These pages are called the statement pack, because most gold evidence is on them.

Retrieval. Code, not the model, picks the pages: the statement pack, the top BM25 pages and the top pages by embedding similarity (Qwen3-Embedding-8B), about 12 pages, from the filing named in the question. This finds a gold page for 87% of the dev questions. Four retrieval options were compared on the dev questions, with a selection rule written down before the results.

Generation. Three LLM calls, with Python checks between them.

1. Call 1 (answering model) returns JSON, with the schema enforced by vLLM: each value it needs, with page, exact quote, column, period and unit, plus a formula over value names. It writes no answer yet.
2. Python checks that each quote is on its page and contains the value, and that the formula uses only value names and six functions. Python then does the arithmetic.
3. Call 2 (same model) sees only the checked values and the result, and writes the answer, citing values as [v1]. Python checks that every number in it is a checked value or the result.
4. Call 3 is a verifier from another model family (Gemma 4 12B). It looks at images of the cited pages and confirms period, unit, column and support.
5. Code attaches the provenance (filing, page, quote, passage). The model never writes a page number.

A failed check gets one retry with the error list, then a refusal. When no filing is named, a router picks up to three candidate filings, each runs the same pipeline, and disagreement returns AMBIGUOUS.

Models ran with vLLM on Georgia Tech's ICE cluster. The pipeline was tuned on 51 dev questions, and the other 99 were run once at the end. Numeric answers were graded by code at the gold answer's precision. Text answers

were graded by gpt-oss-20b, which agreed with the FinanceBench paper's 2,400 human labels 77% of the time.

2. Answers with provenance

Every answer and refusal of every configuration, with page, quote, passage and an image of the cited page, is in the demo and in the repository's results folder. Totals over the 150 questions:

Configuration	Correct	Wrong	Refused
2B, bare	77	72	1
2B, full pipeline	12	2	136
9B, bare	93	57	0
9B, full pipeline	72	16	62
27B (FP8), bare	101	48	1
27B (FP8), full pipeline	80	20	50
27B (FP8), full pipeline, no filing named	76	16	58
GPT-4 Turbo, paper, single store (human labels)	75	17	58
GPT-4 Turbo, paper, shared store, no filing named	29	20	101

Three rows from the 9B answer file:

- *Which of JPM's segments had the highest net income in 2022 Q2?* Corporate & Investment Bank, $3,725 million, computed as the max of four segment values. Evidence: JPMORGAN_2022Q2_10Q, page 27, "Net income | $3,725", plus pages 22, 33 and 37 for the other segments.
- *Does American Water Works have positive working capital in FY2022?* No: $1,250 million current assets minus $2,811 million current liabilities is -$1,561 million. Evidence: AMERICANWATERWORKS_2022_10K, page 81 "Total current assets | 1,250" and page 82 "Total current liabilities | 2,811".
- *Is Verizon's liquidity healthy based on its quick ratio for FY2022?* Refused. Reason: the model found the inputs on page 56 but could not tell which items the quick ratio should include.

3. What worked, what didn't, and what would change

What worked. The checks. For every size they cut wrong answers sharply (9B: 57 to 16), at the cost of correct answers (93 to 72). On 290 questions paired with a filing that holds no answer, the full pipeline gave 0 false answers with 2B and 9B and 1 with 27B. Without the checks the same models gave 53, 20 and 22. Without a named filing, 27B got 76 right, against 29 for GPT-4 Turbo in the paper's matching setup (the graders differ, so this is approximate). Because code does the arithmetic and attaches the citations, no answer can cite text that is not on its page.

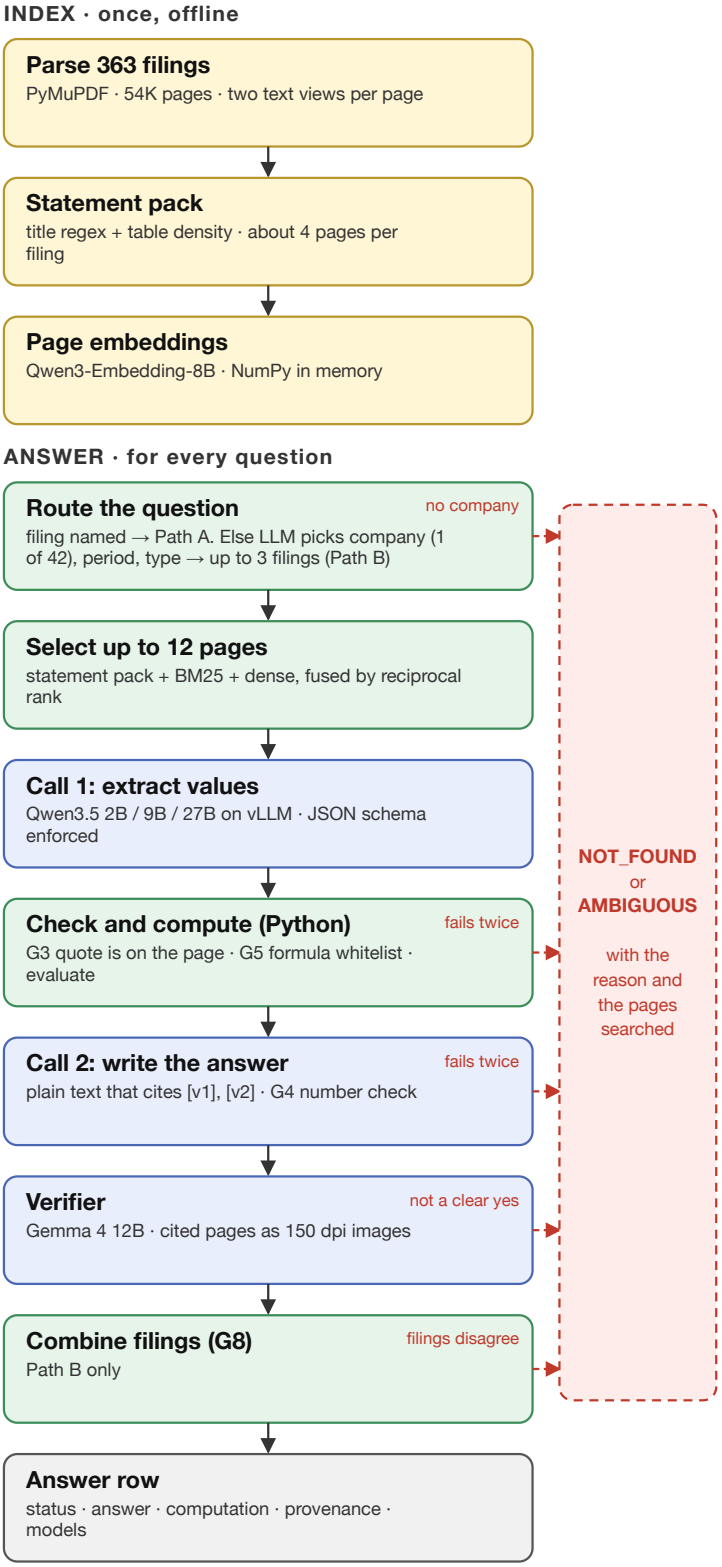
What didn't. The verifier caught only about a quarter of the wrong answers that passed the code checks (7 of 25 for 9B, 4 of 23 for 27B). Most remaining errors use a real, correctly cited number for a different definition of the metric, such as working capital from total instead of operating items. A glossary in the prompt did not help. Size cuts both ways: 27B gets the most correct answers in every setting, but it also passes the checks more often, so more of its errors reach the weak verifier. With the full pipeline it has 4 more wrong answers than 9B on all 150 questions, and 1 fewer on the 99 held-out ones. Most refusals come from the checks, not the model: 9B fails a check twice on 35 questions, often over format. 2B refuses 91% of questions.

What would change, and how to choose the model. For any model: a verifier that also checks the formula against the question, and an answer that states which definition of a metric it used. For the choice: if wrong answers must be rare, 9B or 27B with the pipeline and a stronger verifier. If coverage matters more, 27B. If cost matters, 9B, which is about 3 times faster. Limited compute meant 27B in FP8, a 12B verifier, a 20B judge and one run per configuration. More compute would allow full-precision and larger models, a larger verifier and judge, and repeated runs.

How a question becomes an answer with provenance

Left: each stage and its technology. Right: the decision made at that stage and why. Red: where the system refuses.

data / index · code (deterministic) · LLM call · output · refusal path



DECISION AND REASON

Chunk = one PDF page, not a token window. Tables stay whole, and the page number is the provenance. Plain text feeds search. Layout text (columns joined by "|") is what the model reads.

Most gold evidence is on the income statement, balance sheet or cash flow statement, so these pages are always shown to the model.

No vector database: 54K pages fit in memory. BM25 is built per filing at question time (largest filing: 667 pages).

DECISION AND REASON

Path B keeps up to 3 candidate filings instead of guessing one. Each runs alone, so two filings never share a prompt.

Picked on dev with a rule fixed in advance: the cheapest option within 5 points of the best. Gold-page recall 84% → 87% with dense. LLM query planning added cost, not recall.

Each value comes with its page, exact quote, column, period and scale, plus a formula over value names. The model picks the numbers. It never does arithmetic.

Formulas may use only value names and avg, sum, cagr, abs, max, min. Constants only if they appear in the question. A failed check gets one retry with the error.

Every number in the text must be a checked value or the computed result. One retry with the error.

A different model family, so it does not share the answering model's mistakes. Checks period, unit, column and support. Anything but a clear yes is a refusal.

One filing answers → that answer. All agree → answer. They disagree → AMBIGUOUS, with each value and its source.

Code attaches the provenance from checked values: filing, PDF page, quote, passage, column, scale. The LLM never writes a page number. No checked citation, no answer.

Evaluation sits outside the system and never feeds it: code grades numeric answers and gpt-oss-20b grades text answers against the gold answer. 51 dev questions for every choice above, 99 test questions run once.